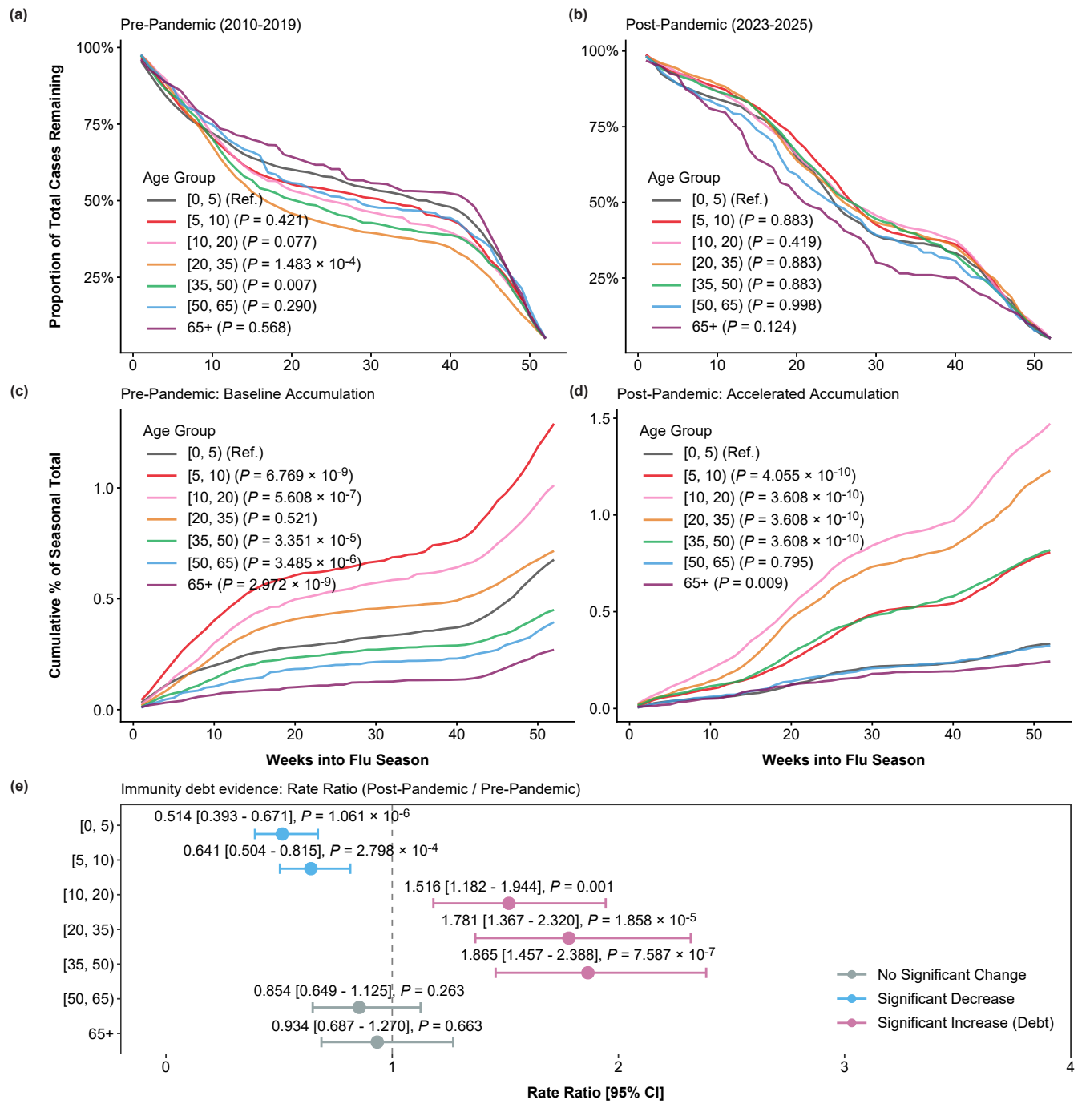

(a)
Pre-Pandemic (2010-2019)
Proportion of Total Cases Remaining
100%
75%
50%
25%
0
10
20
30
40
50
Age Group
[0, 5) (Ref.)
[5, 10) (P = 0.421)
[10, 20) (P = 0.077)
[20, 35) (P = 1.483 × 10-4)
[35, 50) (P = 0.007)
[50, 65) (P = 0.290)
65+ (P = 0.568)
(b)
Post-Pandemic (2023-2025)
Age Group
[0, 5) (Ref.)
[5, 10) (P = 0.883)
[10, 20) (P = 0.419)
[20, 35) (P = 0.883)
[35, 50) (P = 0.883)
[50, 65) (P = 0.998)
65+ (P = 0.124)
(c)
Pre-Pandemic: Baseline Accumulation
Cumulative % of Seasonal Total
1.0
0.5
0.0
Age Group
[0, 5) (Ref.)
[5, 10) (P = 6.769 × 10-9)
[10, 20) (P = 5.608 × 10-7)
[20, 35) (P = 0.521)
[35, 50) (P = 3.351 × 10-5)
[50, 65) (P = 3.485 × 10-6)
65+ (P = 2.972 × 10-9)
Weeks into Flu Season
(d)
Post-Pandemic: Accelerated Accumulation
1.5
1.0
0.5
0.0
Age Group
[0, 5) (Ref.)
[5, 10) (P = 4.055 × 10-10)
[10, 20) (P = 3.608 × 10-10)
[20, 35) (P = 3.608 × 10-10)
[35, 50) (P = 3.608 × 10-10)
[50, 65) (P = 0.795)
65+ (P = 0.009)
Weeks into Flu Season
(e)
Immunity debt evidence: Rate Ratio (Post-Pandemic / Pre-Pandemic)
[0, 5)
[5, 10)
[10, 20)
[20, 35)
[35, 50)
[50, 65)
65+
0.514 [0.393 - 0.671], P = 1.061 × 10-6
0.641 [0.504 - 0.815], P = 2.798 × 10-4
1.516 [1.182 - 1.944], P = 0.001
1.781 [1.367 - 2.320], P = 1.858 × 10-5
1.865 [1.457 - 2.388], P = 7.587 × 10-7
0.854 [0.649 - 1.125], P = 0.263
0.934 [0.687 - 1.270], P = 0.663
No Significant Change
Significant Decrease
Significant Increase (Debt)
0
1
2
3
4
Rate Ratio [95% CI]